

Yun-Chung (Eric) Liu

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EDUCATION

Cornell University, Cornell Tech, Jacobs Technion-Cornell Dual Master of Science

New York, NY

M.S. in Information Systems, Connective Media, May 2028

Washington University in St. Louis, School of Engineering

St. Louis, MO

B.S. in Computer Science, Minor in Human-Computer Interaction, May 2023

Major GPA: 3.79/4.00, Cumulative GPA: 3.72/4.00

Honors: Dean's List (7 consecutive semesters: Spring 2020 - Fall 2022)

SKILLS

Programming Languages: Python, Java, C++, JavaScript, SQL, C#, HTML/CSS

Data & ML/AI: TensorFlow, scikit-learn, Pandas, NumPy, Matplotlib, SQL, Jupyter, Excel

Frameworks & Tools: Spring Boot, ReactJS, Vue.js, Hibernate, Material UI, Git, Jest, Unity

Development Tools: IntelliJ, VS Code, Postman, DBeaver, Docker

PROFESSIONAL EXPERIENCE

Morningstar Inc., *Software Engineer*, July 2023 - June 2026

Chicago, IL

- Led development of Historical Bulk Delivery system, designing scalable Hibernate endpoints and intuitive UI, enabling streamlined file delivery to 1,000+ clients and reducing data input time from days to minutes
- Built Portfolio Observability page to surface key portfolio details including index variants, constituents, hedge data, and FX rates for improved data visibility with backend endpoints for data reload and recalculation, reducing debugging time for operations team
- Implemented SmartX data delivery feature using Spring Boot and HCL, generating \$50K in annual revenue
- Created structured table to organize and display index documents on Morningstar Indexes webpage, indexes.morningstar.com/resources, consolidating 200+ documents for client self-service access

LB Networks, *Software UX Developer Intern*, June 2022 - September 2022

St. Louis, MO

- Designed and improved OcularIP 6.0.4 website functionalities serving 130 service provider companies
- Enhanced Subscription Panel interface by redesigning graph visualizations for improved data clarity and refining button components using ReactJS and Material UI

Ford, *Data Science Intern*, June 2021 - August 2021

Shanghai, CN

- Ran SQL correlation analysis on 100,000+ prospective customers' data (age, showroom visits, number of test drives) to identify top 3 factors driving conversion from prospects to actual Lincoln customers
- Built a visualization dashboard using ReactJS to communicate findings to Ford Lincoln project managers to optimize resource allocation and increase market efficiency

PROJECTS

Manim Lesson Builder, *Independent Project*, September 2026

- Built a pipeline that turns a single structured lesson file into a finished teaching video, automatically producing the animated visuals, narration, captions, and transcript using Python, Manim, and FFmpeg
- Designed the layout and typography engine, added AWS Polly text-to-speech narration, and ran the renderer in a locked-down Docker sandbox isolated from the network and host machine
- Cut cost per 3-minute lesson to about \$0.08, over 200x cheaper than the \$18-\$135 generative video models charge for the same length, which also cannot reliably render correct equations

Footnote, *Independent Project*, September 2026

- Built grounded question answering over SEC EDGAR filings in Python: table-aware chunking and hybrid retrieval combining a hand-written BM25 with vector search via reciprocal rank fusion - no RAG framework, Chroma the only retrieval dependency

FridayFlicks, *Independent Project*, September 2024 - June 2025

- Built an Android movie reviewer app to generate movie ratings based on users' spoken reviews
- Integrated Google Assistant Service for voice-to-text conversion, then used TensorFlow Lite neural network to perform sentiment analysis and generate numerical ratings for user submission

WashU Bear Run, *Team of 4*, Spring 2022

- Led team in developing arcade game from concept to deployment using Unity and C#; coordinated design decisions, task delegation, and integration of obstacle dodging mechanics, power-ups, and scoring system

LEADERSHIP

Cohort Representative, **Connective Media**, *Cornell Tech Student Government*, Aug 2026 - Present

- Elected by peers to represent the Connective Media cohort (70+ students) to program leadership and faculty, advocating for student needs and building a structured alumni mentorship pipeline connecting students with recent graduates in industry

Alpha Kappa Psi Business Fraternity, *Member*, Spring 2020 - Spring 2023